Cultural and Global Influence on Film Industry Revenues using Bass Diffusion Model

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Introduction

This research uses the Bass Diffusion Model to explain how cultural and global factors have impacted the profit revenue of the Hollywood filmmaking industry over the last two decades. By incorporating the innovation effect (p), which represents the influence of advertising and external marketing, and the imitation effect (q), which reflects word-of-mouth and social influence, the model provides a comprehensive framework for analyzing audience behavior. By examining data from nine movie examples, we can observe how advertising methods and cultural influences affect U.S. box office ticket sales. Additionally, by analyzing data from the ten highest-grossing movies during the same period, we can assess how the global COVID-19 pandemic impacted box office performance.

Rationale

I am a movie buff who enjoys many genres and aspire to create my own films in the future. For this project, I aim to demonstrate how advanced mathematical methods can be utilized to understand the art, specifically, the movie entertainment industry.

MATHEMATHICS

Methodology

I classified various movie genres into three types. For Type 1 graph, I grouped Sci-Fi, Horror, and Fantasy. For Type 2 graph, I grouped Romance, Drama, and Historical. For Type 3 graph, I grouped Action, Adventure, and Comedy. One recent box office hit per genre is represented in each graph. Then the Bass Diffusion Model and its Discrete Model form were applied and integrated to describe the weekly fluctuations in movie film sales over a period. Also, the BDM interpreted how various cultural factors and circumstances affected those movies at that time, either through innovation 'p' (influence of advertising, media, initial hype) or imitation 'q' (influence of word-of-mouth, social interactions). As for global factors and circumstances, data on the top 10 highest grossing movies in the last two decades were gathered to determine if any drastic trendline effects could have impacted ticket sales during the release year.

Findings

Bass Diffusion Equation

$$\frac{dF(t)}{dt} = (p + qF(t))(1 - F(t))$$

where:

- $F(t)$ = fraction of the potential market that has adopted (seen the movie) at time t.
- p = **coefficient of innovation** (influence of advertising, media, initial hype).
- q = **coefficient of imitation** (influence of word-of-mouth, social interactions).

$$F_{t+1} = F_t + (p + qF_t)(1 - F_t)$$

or for weekly revenue:

$$\hat{R}_t = m(F_t - F_{t-1})$$

where

- F_t is cumulative adoption at week t,
- m is total market size (or total gross),
- p, q are innovation/imitation.

$$\int \frac{dF}{(p+qF)(1-F)} = \int 1dt$$

$$\int 1dt = \int \frac{A}{p+qF} + \frac{B}{1-F} dF$$

$$A(1 - F) + B(p + qF) = 1$$

$$A - AF + Bp + BqF = 1 + 0F$$

$$A + Bp = 1$$

$$-AF + BqF => F(-A + Bq) = 0F => -A + Bq = 0 => A = Bq$$

$$Bq + Bp = 1 => B = \frac{1}{p+q} \text{ and } A = \frac{q}{p+q}$$

$$\int dt = \int \frac{\frac{q}{p+q}}{p+qF} dF + \int \frac{\frac{1}{p+q}}{1-F} dF$$

$$\text{let } u = p + qF; du = qdF => dF = \frac{du}{q} \text{ and let } w = 1 - F; dw = -dF => dF = -dw$$

$$\int dt = \int \frac{\frac{q}{p+q}}{u} \frac{du}{q} + \int \frac{\frac{1}{p+q}}{w} - dw$$

$$t + C = \frac{1}{p+q} \ln|u| - \frac{1}{p+q} \ln|w|$$

$$t + C = \frac{1}{p+q} \ln|p + qF| - \frac{1}{p+q} \ln|1 - F| = \frac{1}{p+q} (\ln(p + qF) - \ln(1 - F))$$

$$t + C = \frac{1}{p+q} \ln\left(\frac{p+qF}{1-F}\right)$$

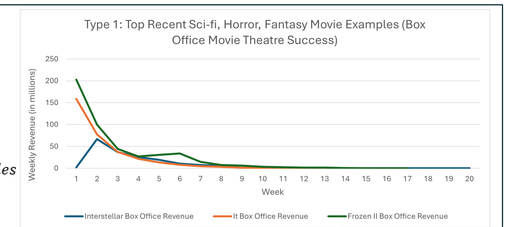
$$t(p + q) + C = \ln\left(\frac{p+qF}{1-F}\right); \textit{take exponential of both sides}$$

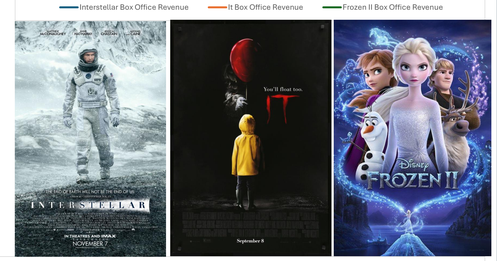
$$Ke^{(p+q)t} = \frac{p+qF}{1-F}; \textit{where } e^C = K$$

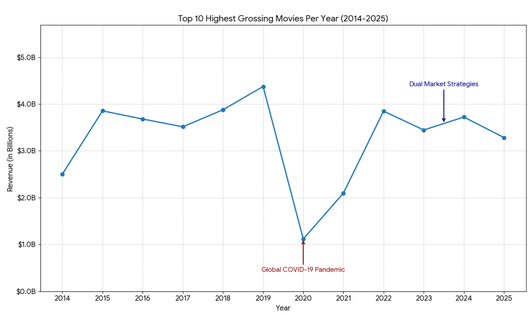
$$Ke^{(p+q)t} - FKe^{(p+q)t} = p + qF$$

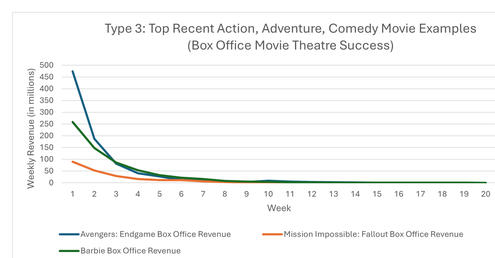
$$Ke^{(p+q)t} - p = F(q + Ke^{(p+q)t})$$

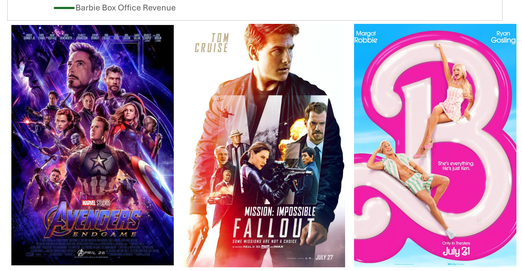
$$F(t) = \left(\frac{Ke^{(p+q)t} - p}{q + Ke^{(p+q)t}}\right)$$

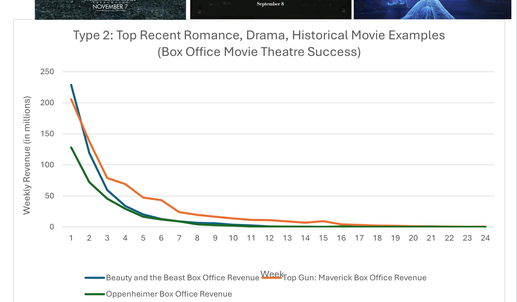


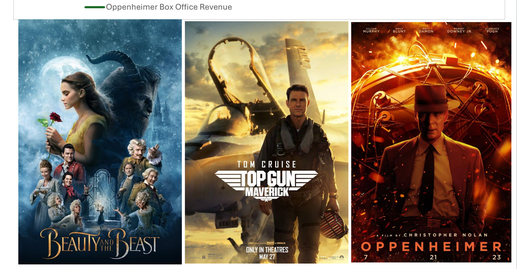












Movie: Interstellar	p=q	p>q	p<q
Total Relative Error	High fluctuation	Small	Large

Movie: It	p=q	p>q	p<q
Total Relative Error	Largest	Larger	Smallest

Movie: Frozen II	p=q	p>q	p<q
Total Relative Error	Hard to determine	Hard to determine	Hard to determine

Movie: Beauty and the Beast	p=q	p>q	p<q
Total Relative Error	Gets larger	Slightly smaller	Slightly larger

Movie: Top Gun: Maverick	p=q	p>q	p<q
Total Relative Error	High fluctuation	Slightly larger	Slightly smaller

Movie: Oppenheimer	p=q	p>q	p<q
Total Relative Error	High fluctuation	Small	Large

Movie: Avengers: Endgame	p=q	p>q	p<q
Total Relative Error	High fluctuation	Slightly smaller	Slightly larger

Movie: Mission Impossible: Fallout	p=q	p>q	p<q
Total Relative Error	High fluctuation	Smallest	Very large

Movie: Barbie	p=q	p>q	p<q
Total Relative Error	High fluctuation	Smallest	Very large

Conclusion

After using the Bass model, and due to the small total relative error, the innovation effect (p) is greater than the imitation effect (q) most of the time, a finding confirmed by the research of Marshall et al. Based on the nine recent movie examples from the three types, the range for p is from .03 to .5 and the range for q is from .01 to .4. Additionally, using the Discrete Bass Model implies that many of these movies drew large crowds through advertising and hyped fans of movie/franchise genres. For films like Top Gun: Maverick and It, the total relative error for q is likely smaller than p because these movies have deep-rooted nostalgia through the return of iconic actors and established fanbases, e.g., Stephen King's literary legacy. These pre-existing connections drive powerful word-of-mouth persuasion that makes social influence a more predictable factor that outweighs external marketing methods. We can also look at the trendline for the top 10 highest grossing movies over recent years where there's a big dip at the year 2020, which was evidently due to the global COVID-19 Pandemic and the quarantine regulations that followed. An article published by Volchenkov et al. also confirms that due to the COVID-19 pandemic, movie theatre ticket sales around the world "plummeted" dramatically in 2020.

Future Work

Future work would involve using the Bass Model to predict future movie revenues, analyze the type of movie genres currently trending, and strategies to support long-term sustainability of the movie theatre industry.

Resources

Marshall, P., Dockendorff, M., & Ibáñez, S. (2013). A forecasting system for movie attendance. *Journal of Business Research*, *66*(10), 1800–1806.
Volchenkov, W., Bichard, S., & Bowman, N. (2022). Covid-19 And Movie Theaters. *Journal of Student Research (Houston, Tex.)*, *11*(4).
IMDb Box Office Mojo (2026)